Phishing Detection & Awareness Report

1. Introduction

Phishing is a type of cyberattack in which attackers send deceptive emails to trick people into revealing sensitive information, clicking malicious links, downloading harmful attachments, or making unauthorized payments.

Phishing emails may use urgent messages, misleading sender addresses, suspicious links, generic greetings, or threats to make recipients act without checking the message carefully.

This project examines three fictional email examples to identify possible phishing indicators, classify their risk, and explain how users can protect themselves. The examples are for educational purposes and are not verified real-world emails.

2. Objectives

The main objectives of this project are:

1. Identify common phishing indicators in email messages.
2. Examine sender information, suspicious links, urgent language, and requests for sensitive information where evidence is available.
3. Classify email examples as Safe, Suspicious, or Phishing based on the available evidence.
4. Explain phishing risks in simple language.
5. Recommend practical ways to recognize, avoid, and report phishing emails.
6. Promote safe email habits and cybersecurity awareness.

3. Scope and Limitations

This project uses three fictional email examples created for training and awareness. The analysis is based on the text provided in each example.

No real email headers, verified sender addresses, attachments, or live email links were provided. Therefore, the examples cannot be treated as verified real-world emails, and their classifications are preliminary.

The project focuses on recognizing warning signs and recommending safe practices rather than proving that a real email account or website is malicious.

4. Methodology and Tools

4.1 Methodology

The following steps were used to organize the email analysis:

1. Prepare three fictional email examples for training purposes.
2. Examine each email's subject line and message body.
3. Identify warning signs such as urgency, generic greetings, suspicious-looking URLs, and unexpected payment requests.
4. Assign a preliminary risk classification based on the available text.
5. Record evidence limitations and avoid making unsupported claims.
6. Prepare practical prevention and awareness recommendations.

4.2 Tools Used

- Google Docs: To prepare the report.
- Notepad: To create and store the fictional email samples as text files.
- Google Admin Toolbox Messageheader: Opened to review the purpose of an email header analysis tool.

4.3 Header Analysis Limitation

The Google Admin Toolbox Messageheader tool requires actual email header information. The fictional examples created for this project do not contain raw email headers. Therefore, no header analysis results, sender authentication results, or delivery-route findings are claimed in this report.

5. Email Sample Analysis

5.1 Sample A — Urgent Account Lock Warning

Subject: Urgent: Your Account Will Be Locked

Classification: Phishing indicators present — treat as potentially phishing.

Indicators identified:

- **Urgency and fear:** The email threatens permanent account lock within 24 hours.
- **Generic greeting:** It begins with “Dear User” instead of identifying the recipient.
- **Request for immediate verification:** It asks the recipient to verify account details immediately.
- **Suspicious-looking URL:** The displayed domain, `secure-account-verify[.]com`, appears designed to look like an account-security service. The domain has not been verified.

Risk explanation:

The message pressures the recipient to act quickly and directs them toward a suspicious-looking link. A recipient could disclose sensitive information if they trusted the message without checking it.

Recommended action:

Do not click the link or enter personal information. Visit the service's official website independently or contact its official support team.

Evidence limitation: The sender address and raw email headers were not supplied.

5.2 Sample B — Unexpected Invoice Email

Subject: Invoice Payment Pending — Action Required

Classification: Suspicious — further verification is required.

Indicators identified:

- The email refers to an invoice that may not be expected.
- It mentions pending payment without naming the supplier.
- The sender's identity and any attachment details are unavailable.

Risk explanation:

Unexpected invoice messages can be used to trick recipients into making fraudulent payments or opening harmful attachments. However, the text alone does not establish that this example is malicious.

Recommended action:

Check whether the purchase is genuine. Verify payment requests using a known supplier contact method. Do not open unexpected attachments until they have been checked.

Evidence limitation: This is a fictional example. No actual sender, attachment, or email headers were examined.

5.3 Sample C — Routine Meeting Confirmation

Subject: Team Meeting Confirmation — Friday

Classification: Appears Safe based on the provided text only.

Indicators observed:

- The message contains routine meeting information.
- It does not request passwords, one-time passwords (OTPs), or payment.
- It does not include a suspicious link or attachment in the supplied text.

Risk explanation:

The message contains fewer obvious warning signs than the other examples. However, ordinary wording does not prove that an email is legitimate.

Recommended action:

Check the sender's address and confirm the meeting through the usual company calendar. Remain cautious if the actual message contains unexpected links or requests.

Evidence limitation: This is a fictional example. The sender's identity and email headers have not been verified.

6. Risk Classification Summary

The following table summarizes the preliminary classifications of the three fictional email examples.

Sample	Main observation	Classification
Sample A — Account Warning	Urgent threat, generic greeting, suspicious-looking URL	Phishing indicators present

Sample B — Invoice Email	Unexpected payment request; sender details unavailable	Suspicious
Sample C — Meeting Confirmation	Routine message with no obvious warning signs in the supplied text	Appears Safe based on text only

Summary of Findings

Sample A contains several warning signs and should be treated as potentially phishing. Sample B requires additional verification because unexpected invoice messages may be used in payment scams. Sample C has fewer obvious warning signs, but its legitimacy cannot be confirmed from the text alone.

These classifications are preliminary and are based only on the fictional message content. No real-world email or sender was verified.

7. Phishing Prevention and Awareness Guidelines

7.1 How to Recognize Phishing

- **Check the sender:** Carefully inspect the complete email address and domain.
- **Look for urgency:** Be cautious of threats, countdowns, or demands for immediate action.
- **Inspect links:** Check the destination before opening a link. If uncertain, visit the official website directly.
- **Be careful with attachments:** Unexpected invoices, documents, and compressed files may be dangerous.
- **Watch for sensitive requests:** Legitimate organizations should not ask you to disclose your password or OTP through an email link.
- **Check the message context:** Confirm unexpected payments, account alerts, or requests for confidential information independently.
- **Remember:** Correct spelling, familiar logos, and professional language do not guarantee that an email is genuine.

7.2 Employee Do's

- Verify unexpected requests using a trusted contact method.
- Use unique passwords and enable multi-factor authentication.
- Keep your operating system, browser, and security software updated.

- Report suspicious messages to your IT or security team.
- If you accidentally click a suspicious link, report it promptly and follow your organization's incident-response instructions.
- Confirm changes to supplier bank details through an independently verified contact method.

7.3 Employee Don'ts

- Do not click suspicious links or open unexpected attachments.
- Do not share passwords, OTPs, or confidential information in response to an email.
- Do not make urgent payments without proper verification.
- Do not trust a message just because it displays a familiar logo or person's name.
- Do not reply to suspicious emails or use contact details provided only in the questionable message.
- Do not forward suspicious messages to colleagues without following your organization's reporting procedure.

7.4 What to Do When You Receive a Suspicious Email

1. Stop and avoid clicking links or opening attachments.
2. Check the sender and the context of the request.
3. Verify the request through an official website or a known contact method.
4. Report the email using your organization's approved reporting process.
5. If you entered a password or disclosed sensitive information, contact the relevant service or security team immediately and change affected credentials through the official website.

Key message: Stop, check, verify, and report before taking action on an unexpected email.

8. Conclusion

This project explored how phishing emails can use urgency, generic greetings, suspicious-looking links, and unexpected payment requests to influence recipients.

Three fictional email examples were reviewed and given preliminary classifications based on their available text. Sample A showed multiple phishing indicators, Sample B required further verification, and Sample C appeared routine but could not be verified as legitimate.

The project also highlighted the importance of checking sender information, verifying requests independently, avoiding suspicious links and attachments, and reporting questionable emails.

Phishing awareness is an important part of cybersecurity. Careful inspection and safe email habits can reduce the risk of credential theft, financial fraud, and unauthorized access.

9. References and Resources

1. Future Interns — Cyber Security Task 2: Phishing Email Detection & Awareness System. https://futureinterns.com/cyber-security-task-2-2026/
2. Google Admin Toolbox — Messageheader. https://toolbox.googleapps.com/apps/messageheader/
3. MXToolbox — Email Header Analyzer. https://mxtoolbox.com/EmailHeaders.aspx
4. Phishing Email Examples — GitHub study resource. https://github.com/autinerd/phishing-mail-examples
5. Phishing Database — GitHub study resource. https://github.com/Phishing-Database/Phishing.Database

Note: The references are resources for the project. The fictional email examples in this report were created for training and were not copied from or verified against these repositories.